One Picture to Study One Thousand Words: Visualization for Qualitative Research in the Age of Digitalization

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INTRODUCTION

Today's world is characterized by an ever-increasing influx of data generated at an unparalleled scale and speed. Trace data resulting from the digitalization of social and organizational lives provides new opportunities for researchers in the social sciences, management, and humanities (George *et al.* 2016; Latour 2010; Rai 2016). By applying computational techniques from statistics, machine learning, and pattern recognition on data that is large in scale, rich in details, and expanding over time, it is possible to understand patterns of individual and group behavior that were once before beyond the reach of the scientific method (Abbasi *et al.* 2016; Agarwal & Dhar 2014; Berente *et al.* 2019; DiMaggio 2015; Johnson *et al.* 2019; Lazer *et al.* 2009; Watts 2013).

While the availability of trace data and the advancement of computational methods to analyze it allow researchers to test new hypotheses and validate theories, the exploration and inductive understanding of social phenomena are more challenging and require new tools and apparatus (Gaskin *et al.* 2014; Pentland *et al.* 2020). This is particularly more prevalent when questions are not well defined and data are unstructured, making quantification and computing less amenable. Researchers trained in the qualitative tradition are familiar with the difficulties, challenges, and efforts involved in gaining a deep understanding of qualitative data (Bansal *et al.* 2018; Finlay 2002). These challenges are only exacerbated when analyzing trace data in digitalized contexts such as in social media, and virtual worlds (McKenna *et al.* 2017; Schultze 2012).

Advances in scientific knowledge are often the synergistic outcomes of advances in methods and theories (e.g., Van de Ven & Poole 1990). For instance, a survey of prior Nobel awards in physics, chemistry, and medicine shows that many winners were recognized for advancing scientific tools and apparatus which in turn inspired advances in theoretical knowledge (Greenwald 2012).

Advancing methods and tools is nothing novel to the qualitative research community that pioneered multiple methods including among others grounded theory, case studies, and process theorization to legitimize qualitative research in the social sciences (Eisenhardt 1989; Glaser & Strauss 1967; Langley 1999; Lee 1989). Qualitative researchers in the information systems research community also pioneered advances in qualitative methods including mixed-methods, processes research, and computational theory development (Berente *et al.* 2019; Gaskin *et al.* 2014; Pentland *et al.* 2020; Venkatesh *et al.* 2013).

In this chapter, we argue for further advancing qualitative research methods by creating tools to investigate digital traces of digital phenomena. We specifically focus on large-scale textual data sets and show how interactive visualization can be used to augment qualitative researchers' capabilities to theorize from trace data. We ground our approach on prior work in sensemaking, visual analytics, and interactive visualization (Munzner 2014; Russell *et al.* 1993) and show how tasks enabled by visualization systems can be synergistically integrated with the qualitative research process. Finally, we apply these principles with several open-source text mining and interactive visualization systems.

We hope that this chapter stimulates further interest and provides specific guidelines for developing and expanding the repertoire of open-source systems for qualitative research.

QUALITATIVE RESEARCH METHODS IN THE AGE OF DIGITALIZATION

We use the term digitalization to refer to the increased prevalence of information and communication technology as a medium for social activities and interactions. Digitalization is pervasive in both the personal and professional lives of many people worldwide. Digital technology is unique in that it leaves a digital exhaust of trace data as a byproduct of online interactions and activities (Neef 2014). While acknowledging the privacy concerns of making social activities largely transparent (Stohl *et al.* 2016; Zuboff 2015), trace data of digitalized contexts offer a unique opportunity for researchers to study and understand social phenomena at increasing scale and scope (Berente *et al.* 2019; Latour 2010).

Researchers using qualitative research methods face two special challenges when examining trace data. First, the scale of the data challenges the capacity of researchers to systematically, comprehensively, and rigorously analyze all data (Gioia *et al.* 2013). Second, because digital trace data is typically left data, researchers are distant from the focal phenomenon, which challenges their ability for the iterative and reflexive sensemaking (Finlay 2002; Walsh 1995) and the degree to which they can bring their background knowledge and personal experience in the process of knowledge creation (Cook & Brown 1999; Polanyi 1962; Tsoukas 2009). These two challenges are interdependent. For instance, the grounded theory methodology suggests that researchers build an incremental understanding of data to generate theory and augment their data until they reach theoretical saturation (Glaser & Strauss 1967; Walsh *et al.* 2015).

Recently, there have been multiple attempts to bridge the gap between qualitative and quantitative research. These approaches fall within the broad category of mixed-methods and aim to provide guidelines and best practices for integrating quantitative and qualitative methods for studying quantitative and qualitative data (Berente *et al.* 2019; Lindberg 2020; Venkatesh *et al.* 2013; Zachariadis *et al.* 2013). Mixed-methods have been successfully applied in studying multiple digital phenomena including online communities (Levina & Vaast 2015), digital platforms (Hukal *et al.* 2020), and social media (Vaast *et al.* 2017). However, whereas mixed-methods bridge the gap by substituting qualitative inquiry for quantitative inquiry, they do not expand the scope for qualitative methods to study digital trace data.

Focusing on this goal, we maintain that new apparatuses are needed to study digital phenomena (Østerlund *et al.* 2020). As an illustration of such an apparatus, consider ThreadNet an open-source tool for contextualizing events using networks and visualizations (Pentland 2016). While digital traces provide rich and fine-grained data about processes, contextual information is often missing. By enriching it with context, ThreadNet makes process trace data more informative for systematic inquiry about the dynamics of processual phenomena (Pentland *et al.* 2020).

Within the overall objective of creating tools for advancing qualitative research methods, we focus on tools that facilitate the study of a large set of textual documents from trace data. Notwithstanding the importance of different kinds of trace data and different techniques for analyzing it (Berente *et al.* 2019), researchers' understanding of data underlying the phenomenon is the cornerstone of qualitative research (Bansal *et al.* 2018) and is most challenging when data is large and distant from the researcher. Information systems for improving sensemaking can bridge this gap between digital trace data and qualitative researchers. We outline below the principles of designing such systems and the predominant role of interactive visualization in enabling sensemaking tasks.

INTERACTIVE VISUALIZATION FOR IMPROVING HUMAN SENSEMAKING

Humans are *informavores;* they survive in their environment by constantly ingesting and organizing information in a two-process loop (Miller 1984). The first process is *information foraging, which* refers to information seeking, gathering, and consumption. The second process, *sensemaking,* refers to representing information in mental schemas to develop insights through the manipulation of these schemas and the creation of knowledge or direct action based on the insight (Pirolli & Card 2005; Russell *et al.* 1993). A better understanding of information foraging and sensemaking allows people, when possible, to modify their strategies or the structure of the environment to maximize their rate of gaining valuable information (Pirolli & Card 1999).

Information foraging extends to many situations in which people seek to understand issues and events that are novel, ambiguous, confusing, or in some other way violate prior expectations (Maitlis & Christianson 2014). This is particularly true for researchers when novel observations yield surprising results and call for reconsidering existing models and theories (Richardson & Kramer 2006). The need for sensemaking arises when people experience a cognitive gap resulting from novel observations (Dervin 2003). Sensemaking involves fitting new data into existing mental models and adjusting mental models around data (Klein *et al.* 2006; Klein & Moon 2006).

Effective analytical reasoning requires a mix of information foraging and sensemaking (Shrinivasan & van Wijk 2008). For example, qualitative researchers working with text documents need to connect threads of evidence to create concepts and categories for understanding the body of texts (Gioia *et al.* 2013; Langley 1999). Researchers can do this manually for a small number of documents, but as the volume grows and the corresponding number of concepts and categories, within the documents grow larger, foraging and sensemaking processes become difficult for manual analysis (Stasko *et al.* 2008). In the past three decades, computer and information systems have been designed to improve and support information foraging and sensemaking.

Improving Information Foraging

Information foraging involves collecting and organizing external information. For example, qualitative researchers working with text documents need to connect evidence to form hypotheses and theories of potential interest. This work involves reading and organizing text and identifying concepts and entities in addition to their relationships (e.g. spatial, temporal, causal). The constructed mental models allow for further development of hypotheses in the sensemaking process. The foraging process relies on human memory to process information and represent schemas as internal mental models. Limits of internal cognition and memory limit the scaling of this process of a large amount of data (Neisser 1976).

To overcome the limits of internal cognition and memory, *external representations* of information provide a shareable object of thought and change the cost structure of the inferential landscape (Kirsh 2009). External representations swap cognition with perception by invoking the human visual system to perform a significant amount of visual information processing in parallel with cognition at the preconscious level (Munzner 2014 Chapter 6). External representations can be used to substitute and to complement internal representations in improving cognitive tasks (Hegarty 2004; Liu *et al.* 2008). Herbert Simon argued that mathematics can be viewed as changes in representation that make explicit what was previously implicit: "Solving a problem simply means representing it so as to make the solution transparent" (Simon 1996 p. 132).

Visual external representation is often used because of the superiority of human vision and visual perception of all other senses (Anderson *et al.* 2005). While humans possessed advanced visual

sensory and perception, developing language, particularly reading and writing, came much later. External representations seek to shift the burden between cognition and perception by visualizing textual and numerical data (Munzner 2014). Larkin and Simon (1987) provide several examples of problems that are very hard to solve without relying on external visual representations.

Visual representations are used for the construction of arguments and the exploratory investigation and evaluation of alternative hypotheses (Shrinivasan & van Wijk 2008). Alternative data representations are visually presented to analysts to support analytical reasoning (Endert *et al.* 2014). Early visual analytics systems focused on transforming data to produce multiple representations that evoke different situated cognitive strategies and lead to more efficient and effective reasoning (Liu *et al.* 2008). Visual external representation also offers several benefits such as organizing data by spatial location to speed up search and recognition and bundle similar information for a specific problem-solving inference (Munzner 2014).

Creating visual representations is our first guiding principle for empowering qualitative researchers to investigate trace data of digital phenomena.

Improving Sensemaking

The sensemaking process relies on the schema produced in the information foraging process to make decisions and reach insights. External representation helps to alleviate the cognitive overload of short term memory by engaging the visual perception system to ingest a large volume of information. However, human attention also has severe limits. Conscious search for items is an operation that grows more difficult as the number of items increases (Shirley *et al.* 2015 Chapter 20). Performance in search tasks degrades quickly with far worse results after several hours than in the first few minutes (Ware 2012). Systems designed for improving sensemaking should enable its users' focus on making judgments while automating the routine tasks. This principle is referred to as Human-In-The-Loop (Schirner *et al.* 2013).

The Human-In-The-Loop principle stresses the importance of interactivity between the system and its users and augmenting their cognitive capability rather than replacing it (Endert *et al.* 2014). Textual data sets are hard to represent and visualize in a single static view. By responding to user input and changing the view on demand, the system overcomes the limitations of a static visual encoding and enables the user to comprehend the high-level patterns but also sense make based on the details (Munzner 2014 p. 244). Interactive visualization is different from but complementary to static visualization typically used in reporting research results. Whereas static visualization enables researchers to communicate their understanding of the data ex-post, interactive visualization enables researchers to gain such understanding.

Interactivity is the second guiding principle for empowering qualitative researchers to investigate trace data of digital phenomena.

Zooming in and out from Data to Representation and Back

One key task in analyzing textual data involves synthesizing concepts and ontologies from the text (Gotz *et al.* 2006; Wright *et al.* 2006). Discovering concepts, exploring their relationships, and eliciting evidence from documents is one step toward gaining a comprehensive understanding of the target phenomenon. However, developing a personal understanding of this text requires a deep and reflexive interaction through it (Charmaz 2006 Chapter 8; Finlay 2002) where researchers reflect upon their past experiences as well as experiences of the "generalized other" people represented in the data. New knowledge is created when researchers explicate their background knowledge and take distance from their old understanding of the phenomenon (Tsoukas 2009).

Zooming in and out is used as a metaphor when investigating digitalized phenomena. This metaphor can further guide creating research apparatus. For example, through computational apparatus, complex digital trace data can be examined across multiple spatial and temporal cross-sections via computational methods (Gaskin *et al.* 2014; Latour 2010). In qualitative research, zooming in and out enables forming understanding at multiple levels of abstractions (Gioia *et al.* 2013) as well as connecting the dots between these levels across contexts (Nicolini 2009).

Visual analytics systems embrace the continuous exploration of the data (Chau *et al.* 2011). These systems can facilitate zooming in and out of the data by presenting both the original text and its summary representation (Shrinivasan & van Wijk 2008). Beyond presenting the original data and its visual summary, systems can provide summaries at different levels of granularities with linkages between these levels (Brehmer *et al.* 2014; Stasko *et al.* 2008).

Providing multiple representations of trace data at different levels of granularity along with their associations is the third guiding principle for empowering qualitative researchers to investigate trace data of digital phenomena.

In sum, visualization, interactivity, and multiple representations are the three guiding principles for designing visualization systems for qualitative research of trace data. Surprisingly, qualitative and even quantitative research in social science rarely uses interactive visualization during the research phase. Systems are developed in other disciplines to combine the best of computation and human perception for analyzing textual documents. Building on the three principles, we show how visualization can be used in tandem with qualitative theorizing from data.

INTERACTIVE VISUALIZATION FOR QUALITATIVE RESEARCH

We now show how interactive visualization can be used to improve the process of qualitative research. We focus on grounded theory methodology (GTM) as a class of methods to build a theory based on the systematic analysis of qualitative data (Charmaz 2006; Morse *et al.* 2009; Suddaby 2006; Urquhart 2013; Walsh *et al.* 2015). As outline by Tweed & Charmaz (2011), this particular focus enables us to be specific about the different stages in the research process (Figure 1). Notwithstanding this focus, the association between the research stages and the roles of interactive visualization to improve them is general and should apply to different traditions in qualitative research (Bansal *et al.* 2018).

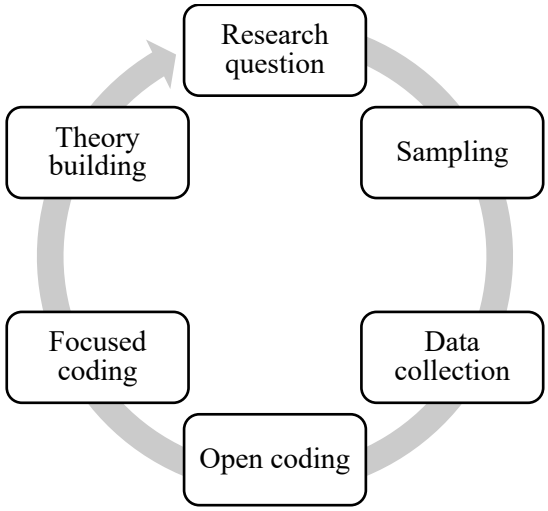

Figure 1: The process of grounded theory methodology

Grounded Theory Methods Research Process and Challenges of Trace

Figure 1 illustrates the process of GTM. Briefly stated, the process begins from an initial research interest followed by theoretical sampling to collect data suitable to answer the question. This data is analyzed incrementally with open coding to identify concepts and categories. More data is added incrementally to expand and add nuances to identified themes until reaching saturation. Next, selective and focused coding is applied to identify several types of relationships among identified concepts. At this stage, defined concepts and relationships are constantly compared to different interpretations of data. These interpretations are finally weaved into an emergent theory, and a chain of evidence is created to transparently show how this emergent theory came about (Charmaz 2006; Glaser & Strauss 1967; Walsh *et al.* 2015). Within this process, four activities are challenging with trace data: theoretical sampling, coding, constant comparison, and the creation of a chain of evidence.

Theoretical Sampling

Theoretical sampling strategies are important in grounded methods because they allow researchers to get into the data and provide a significant direction for theorizing (Thomas & James 2006; Urquhart *et al.* 2010). With trace data, this important aspect of grounded theorizing is transformed into creating questions for the systematic yet meaningful extraction of meaning for the analysis and theory building (Pan & Tan 2011). First, instead of limited access to in-depth data such as interviews and observations, researchers have quasi-unlimited access to distant data. Sampling becomes an issue of reducing the large data set and selecting parts that can lead to interesting insights. Second, the collected data lacks context. Although the data are large, it is fragmented and may not necessarily result in a cohesive understanding of the phenomenon under consideration. Search for leads in the data and reconstructing context involves a huge cognitive overload. As a result, researchers risk gravitating towards specific understandings of multifaceted phenomena rather than considering all potential interpretations offered by data (Levina & Vaast 2015). Theoretical sampling in this context thus challenges the "relatedness" validity criterion, that is, the extracting of a meaningful sample from a large data set.

Coding

Within GTM, the coding process involves two approaches: substantive coding (which typically includes open and selective coding) and theoretical coding. Whereas substantive coding refers to the conceptualization of empirical data (and often is taken directly from the language of the data, via in vivo codes), theoretical coding involves advanced abstractions to provide a framework for enhancing the explanatory power of the emerging grounded theory. As with other concepts in GTM, theoretical codes must show 'earned relevance'; that is, they must not be preconceived or forced into the analysis. Still, it is easy to fall into the trap of conceptualizing concepts that are known from the start. To address this challenge, researchers can use predefined high-level sociological constructs known as "coding families" (Glaser 1978).

Even with these suggestions, theoretical coding remains a source of struggle among grounded theorists (Glaser 2005, 2013). This challenge is only exacerbated by the volume and variety of trace data. There is a lot more data to code and more concepts to explore. Relying on pre-existing codes can be enticing given the large number of documents that researchers need to sift through. Glaser (2013) warned that researchers should apply as many theoretical codes from as many sources as possible. They should also restrain from the temptation of over-relying on one's "pet" codes, that is, those associated with theories researchers happen to be familiar with. In other words, researchers, as they proceed to theoretical coding, should refrain from trying to "fit the data" into

preconceived conceptualization. On the other hand, the families of codes suggested by Glaser (1978) and Strauss and Corbin (1998) may already be too limiting. Thus, "appositeness," which is a validity criterion referring to the suitability or fitting of the data to the context, is challenged in the context of large textual trace data sets.

Constant Comparison

Constant comparison is an analysis technique frequently used within GTM (Urquhart *et al.* 2010). The premise behind this technique is that by comparing and contrasting many similar and dissimilar data units, a researcher is more likely to generate dense categories and properties about the phenomenon under study. For example, a researcher might want to compare all social media users with a certain property value with all other users with different property values and analyze what other properties emerge as salient differences. She or he may also use a "flip-flop" technique (Strauss & Corbin 1998), which involves turning a concept upside down and looking for extreme cases. Another approach to constant comparison is to conduct "far-out" comparisons, where a researcher compares and contrasts a contextualized data set with another data set associated with a different context to potentially uncover new properties that might not have been obvious to the researcher from the onset.

Whereas constant comparison is a powerful approach to increasing one's theoretical sensitivity (Glaser 2005, 2013), it becomes a challenge when applied to trace data. For example, researching social media impact, a researcher has an immense population to leverage. Even a small slice of such potential data set will be quite significant in volume (Vaast *et al.* 2017). The researcher is also constrained by whatever the data reveal. For example, the data may about a single platform or one group of users (Levina & Arriaga 2014), making comparisons with other platforms or groups of users impossible. Constant comparison can benefit from integrating and relating data sets from different sources, although this is not practically feasible given the lack of standardization of data formats. "Comprehensiveness," therefore, is at stake when proceeding to the constant comparison approach in the context of trace data of digital phenomena.

Chain of Evidence

One of the strengths of the grounded theory method is its ability to provide a chain of evidence (Charmaz 2006). Indeed, given the detailed approach and guidance provided by the method, grounded theorists should be in a position of strength to explain their coding process, how categories (and their properties) emerge, how relationships between categories were established, and how categories were combined to form larger concepts. By doing this, they avoid charges often directed to qualitative researchers, i.e., that they are selective about the data they use to back-up their theory (Urquhart 2013). Whereas such a chain of evidence always adds to trustworthiness, it can, in some epistemological traditions such as positivism or critical realism (Zachariadis *et al.* 2013), also be used in verification, wherein other researchers can independently verify the findings by reproducing the grounded theorist's analysis.

Verification and more generally the quality, reproducibility, and accountability of qualitative research have been the subject of much attention in the literature (Seale 1999) not only for grounded theorists but also for all qualitative researchers. Although the chain of evidence can be used to support trustworthiness or encourage verification, it is rarely detailed in journal-length publications, primarily due to space limitation. Grounded theorists, indeed, often have to choose between explaining their theory or the chain of evidence leading to it (Urquhart 2013). If they do expose their chain of evidence, it is often in a highly summarized format, where a subset of raw data is shown to support codes that are then combined into categories. Often missing are the details

of how the researcher revealed categories (e.g. by comparing slices of a data set via the constant comparative technique, discussed earlier), how properties emerge as particularly relevant or irrelevant, and which alternative open coding families were considered, and the insight they provided. Dealing with trace data compounds this challenge, as publication space limitations in addition to the expansiveness of analyzed data impose a trade-off in writing and reporting findings. On the downside, not being as transparent and comprehensive regarding the chain of evidence calls into question the trustworthiness of the research findings.

Table 1 summarizes the key challenges that trace data pose to grounded theorists. The availability of trace data combined with the advances of computational techniques has led to increased interest in computational methods to analyze the data and mixed-methods for combining quantitative and qualitative inquiries. Not addressing these challenges compels researchers to favor quantitative methods over qualitative ones. Fortunately, computational techniques can be used for enriching qualitative research as we describe below.

Table 1: Key challenges to grounded theorists relying on textual trace data

Challenged validity criteria	Challenged research activities	Description of the challenge
Relatedness	Theoretical sampling	Extracting a meaningful sample for a large data set
Appositeness	Theoretical coding	Coding a large number of documents without being limited to a pre-existing set of codes
Comprehensiveness	Constant comparison	Comparing and contrasting data and codes across different dimensions
Trustworthiness	Chain of evidence	Summarizing and presenting results

Interactive Visualization Tasks

We outlined the three guidelines for designing interactive visualization systems to support qualitative research. We now expand on how such a system can meet the challenges of researching trace data. We rely on the taxonomy of tasks enabled by a visualization system illustrated in Figure 2, adapted from (Munzner 2014 p. 47). In the discussion below, we use the term “the system” to shortly refer to the interactive visualization system designed to enable these tasks.

Analyze

The system's most common task is to allow the user to understand existing data presented visually to generate new knowledge. This new knowledge can be a new hypothesis but also a validation or disconfirmation of an existing one. The system can also allow the user to annotate existing data and visualizations to record this new knowledge (Munzner 2014 Chapter 3).

In GTM, researchers usually go through several rounds of coding (typically, open, axial, and theoretical). Open coding is a key analytical task in GTM in which researchers make sense of data, their experiences, and reflect upon meanings and actions. These meanings entail labeling bits of data according to what they indicate for conceptual development (Charmaz 2006 Chapter 5). Presenting high-level summaries of text can help in this step. In axial coding, initial codes are sorted, compared, and synthesized. Multiple visualizations can support different coding tasks (i.e., open, axial). Here, the system can automate the process of creating an initial set of codes, a process that is often extremely tedious and time-consuming, and allow the search for further refinement of codes, as well as annotating the data with these codes. The system can accelerate the open and axial coding so that researchers can focus on theoretical coding.

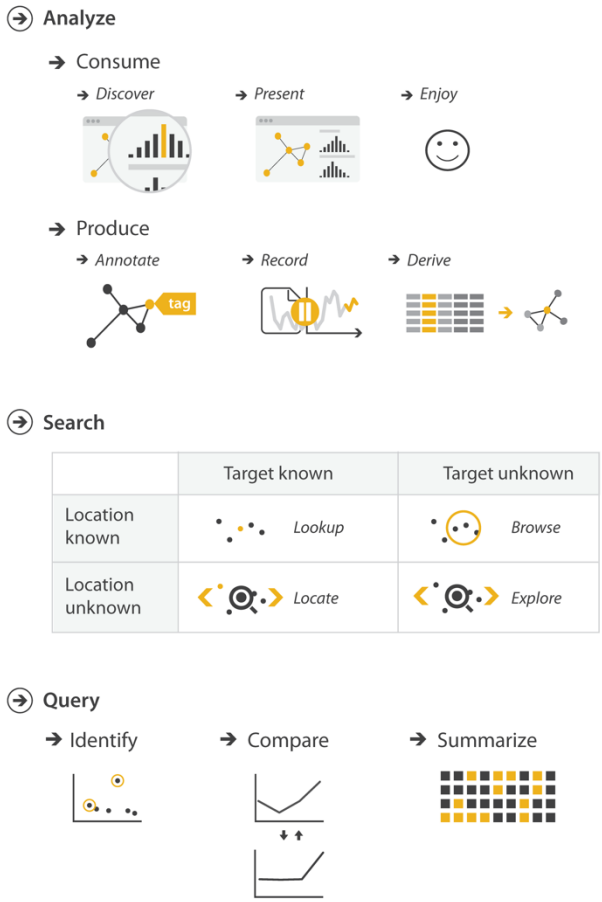

Figure 2: Tasks enabled by visualization, adapted from Munzner (2014 p. 47)

Search

Another common task enabled by the system is searching for information of interest. This task is very important in the context of trace data. The search task can be just a simple lookup if users already know what they are looking for, locate if the data is known but their location is unknown, or browse if the location is known. Finally, if neither data nor their location is known, the system can nonetheless support exploration tasks (Munzner 2014 Chapter 3).

In the context of GTM, searching can be useful in the sampling phase. Traditionally, GTM researchers collect rich data from primary sources. However, with trace data, it becomes important to effectively search for these sources in the heap of available documents. Second, within identified documents, a search is needed to locate interesting excerpts. This is necessary for long documents and can substitute the intensive interviewing techniques employed with primary sources. Searching can be based on codes identified previously, or using semantic criteria to group documents similar to others that the researchers deem interesting.

In addition to identifying documents of interest, the system can link and integrate the documents to reconstruct context. By connecting and linking fragmented unstructured textual data, the system helps researchers deal with establishing the boundaries of their investigations. Hence, link-by-link, researchers can discover new texts of potential relevance. Connecting documents can be automated by identifying common concepts and arranging them over time. Finally, researchers can define texts of primary importance to them to aid the system in reorganizing the searched collection.

Query

Once the target data and location are known, users can query it to identify, compare, and summarize. The progression of these three activities corresponds to an increase in the inclusivity of identified data: one, some, or all. Identify refers to a single target, compare refers to multiple targets, and summarize refers to the full set of possible targets (Munzner 2014 Chapter 3).

These tasks can be used to support theoretical sampling, selective and axial coding, and theory building. Theoretical sampling requires adding further selective data to refine and fill major codes. This can be supported by identifying documents to match existing codes. Coding can be facilitated with direct querying to summarize un-coded documents. This task can be extended to include relationships among codes. Theory-building can benefit from querying for comparison and summaries (e.g. constant comparison). The system can also locate similar and contrasting documents based on a search criterion or starting from an initial set of documents.

Finally, the system can provide shareable visualizations that can be used to promote the transparency and reproducibility of findings. Transparency can be promoted by providing the audience of the research with access to the same interactive visualization that the researcher has. This is especially relevant when trace data is public and ethical and legal issues of sharing access are addressed appropriately. By providing links to the data and a chain of evidence, the reproducibility of the research is improved. Visualization as well as underlying data can serve as an open repository of knowledge that other researchers can use to build cumulative knowledge of the target digitalized phenomenon.

Table 2 summarizes how interactive visualization tasks can address the challenging research activities and validity criteria associated with qualitative research of trace data.

Table 2: How to proposed system implementation responds to challenges of using GTM with big data

Challenged validity criteria	**Challenged research activities**	**Interactive Visualization**
Relatedness	Theoretical sampling	• Creating a high-level presentation of text • Creating an initial set of concepts and providing leads for related documents
Appositeness	Theoretical coding	• Automating the creation of open codes • Experimenting with pre-existing codes • Freeing researchers' time to focus on theoretical coding
Comprehensiveness	Constant comparison	• Integrating and relating data sets • Searching and contrasting many similar and dissimilar data units
Trustworthiness	Chain of evidence	• Sharing access to data visualization with the research audience (transparency) • Promoting the reproducibility of findings

SYSTEM EVALUATION AND DEMONSTRATION

We now demonstrate how visualization tasks can be implemented with open-source systems. For this discussion to generalize to other systems, we first explicate our criteria for selecting these systems as well as the general criteria for evaluating any visualization system.

Selection Criteria

Text visualization is a vast area of study and there are many open-source text visualization systems (Kucher & Kerren 2015, 2019).[i] The systems we chose are useful for social sciences and management because they have been developed with research applications in mind. In selecting the three systems, we first excluded proprietary text mining and visualization systems such as Leximancer (McKenna *et al.* 2017) and MineMyText (Müller *et al.* 2016). We focus mainly on open-source software not only because of its low cost but because openness, transparency, and extensibility are the core ethea of science that need to be promoted in the digital era.

Evaluation Criteria

Algorithms

All visualization systems rely on data processing algorithms for accessing and manipulating the underlying data sets. Visualization algorithms are beyond the scope of this chapter, but they are nonetheless, an important criterion when evaluating a visualization system (Munzner 2014). Also, of particular interest to visualizing trace data are algorithms used to process text. The algorithms fall under the broad umbrella of natural language processing (NLP). Here again, NLP is a large and fast-evolving field (Eisenstein 2019). We outline four mainstream classes of NLP algorithms used in contemporary text visualization systems, including the three systems we review.

First, to provide access to large text, text visualization systems use *information retrieval* algorithms to index the text and provide fast search capabilities (Manning *et al.* 2008). The information retrieval algorithms can also provide a semantic layer from grouping excerpts that are related together (Landauer *et al.* 2013). For example, because the two words "horse" and "rider" come together in many documents, they have short semantic distance suggesting that the term "horse rider" is a standalone entity or concept (Cilibrasi & Vitanyi 2007).

Second, the main challenge with textual data sets is to lend the data amenable to quantification and computation. Given the focus on identifying concepts and themes that map to theoretical codes and extracting relationships to formulate hypotheses, *named entity recognition* (NER) can be used for extracting higher-level representations (e.g., concepts and entities) from text is one central task in NLP (Satoshi Sekine & Nobata 2002). NER algorithms typically focus on categories such as locations, names, organizations, dates, and money (Klein *et al.* 2003). NER systems are built with supervised training using labeled data. State-of-the-art NER systems are very precise reaching almost human-level capabilities (Collobert *et al.* 2000).

Third, although precise and standardized, NER does not work well for concepts that do not fit within its predefined categories. Many often concepts are not pre-defined but rather emergent from data. *Topic modeling* is one unsupervised approach that aims to help text analytics by identifying latent topics based on co-occurring words (Chuang *et al.* 2012). The emergent nature of topics makes them useful for exploring trends in large textual documents such as science publications (Blei & Lafferty 2007). Topic models relying on word co-occurrence in documents have been extended to model authorship (Rosen-Zvi *et al.* 2010), time (Dou *et al.* 2012), and relationships among topics (Blei & Lafferty 2007). Topic modeling has recently reached wide adoption in the social sciences (DiMaggio *et al.* 2013; Hannigan *et al.* 2019).

Fourth, more recently, textual analysis using *deep neural networks* (DNN) became a mainstream approach with many promising results. DNNs benefit from large data sets making them a natural choice for analyzing big unstructured data. DNNs can outperform prior NLP techniques (Collobert *et al.* 2000; Collobert & Weston 2008). Deep neural networks are capable of extracting a higher representation from raw data (LeCun *et al.* 2015). Such representation includes meaning and semantics when data are text (Bengio *et al.* 2013). Word representation using DNNs are used to extract sentiments (Socher *et al.* 2013), syntax (Mikolov *et al.* 2013; Turian *et al.* 2010), and even semantics from the raw unprocessed text (Kumar *et al.* 2015; Xiong *et al.* 2016).

Visual Encoding and Interaction Idioms

Research in information visualization shows that different types of visual channels vary in their effectiveness in human perception. As a result, specific visual channels are recommended for presenting different types of data. For example, categorical data are best represented with shapes and colors. Ordinal data are best visualized with size or position on a scale. Relational data are best represented with networks, proximity, and containment areas (Munzner 2014). Compared to these data types, there is less research on visual idioms for text (Stasko *et al.* 2008). A small amount of text can be presented in verbatim or tabular views. However, this is not feasible for a large amount of text because reading and understanding text is a cognitively taxing process.

Modern text encoding idioms emphasize the importance of words, concepts, and topics. For example, a word cloud provides a high-level summary of text by highlighting the most frequent (Heimerl *et al.* 2014). Another powerful external representation for text is a network representation (Singhal 2012). This representation assumes that if concepts and relationships from the text are known, they can be visualized with a graph. This visualization summarizes the text and allows for comprehending its scope and meaning without reading it. These concepts and relationships can be manually labeled such as in GTM open and axial coding.

Visualization systems supporting multiple representations provide multiple views of the data to support the tasks of analyzing, searching, and querying. The views should be linked to allow the user of comprehending the data at different granularities and minimize complexity when necessary. One common configuration includes views of data, knowledge, and navigation (Shrinivasan & van Wijk 2008). The navigation view allows for searching and filtering with automatic summarization and a high-level presentation of the data set. The knowledge view allows for the selection, exploration, and zooming in and out to specific documents and excerpts. Finally, the data view provides the original textual data as well as the context in which it occurs.

Open-Source Systems for Interactive Visualization of Text

Topic Modeling and Topic Visualization

First, we focus on topic modeling which is a technique gaining wide adoption in social science and management (DiMaggio *et al.* 2013; Grimmer & Stewart 2013; Hannigan *et al.* 2019; Safadi *et al.* 2020; Schmiedel *et al.* 2019). Topic modeling uses the latent Dirichlet allocation (LDA) algorithm (Blei et al. 2003) to categorize the content of many text documents into a set of substantively important coding categories or topics. As an algorithmic approach that requires a minimum of human intervention, the method is more inductive than traditional approaches to text analysis in the social and human sciences (Mohr and Bogdanov 2013). Many open-source packages for topic modeling and visualization are readily available such as MALLET (McCallum 2002), Gensim (Rehurek & Sojka 2010), Termite (Chuang *et al.* 2012), and LDAVis (Sievert & Shirley 2014).

As an example of using topic modeling, consider a researcher examining the mailing list of an open-source community (Safadi 2015 Chapter 6). Topic modeling can be used to uncover common frequent words and their associations with the mainlining list posts. Figure 3 shows such topics using the Termite visualization package. The researcher can use this visualization to understand the themes in the mainlining list, give these topics theoretical labels and discard noisy ones. For example, the induced topics can be grouped to topics about technical tasks (e.g., programming), support tasks (e.g., installation), and community communication (e.g., organizing meetings).

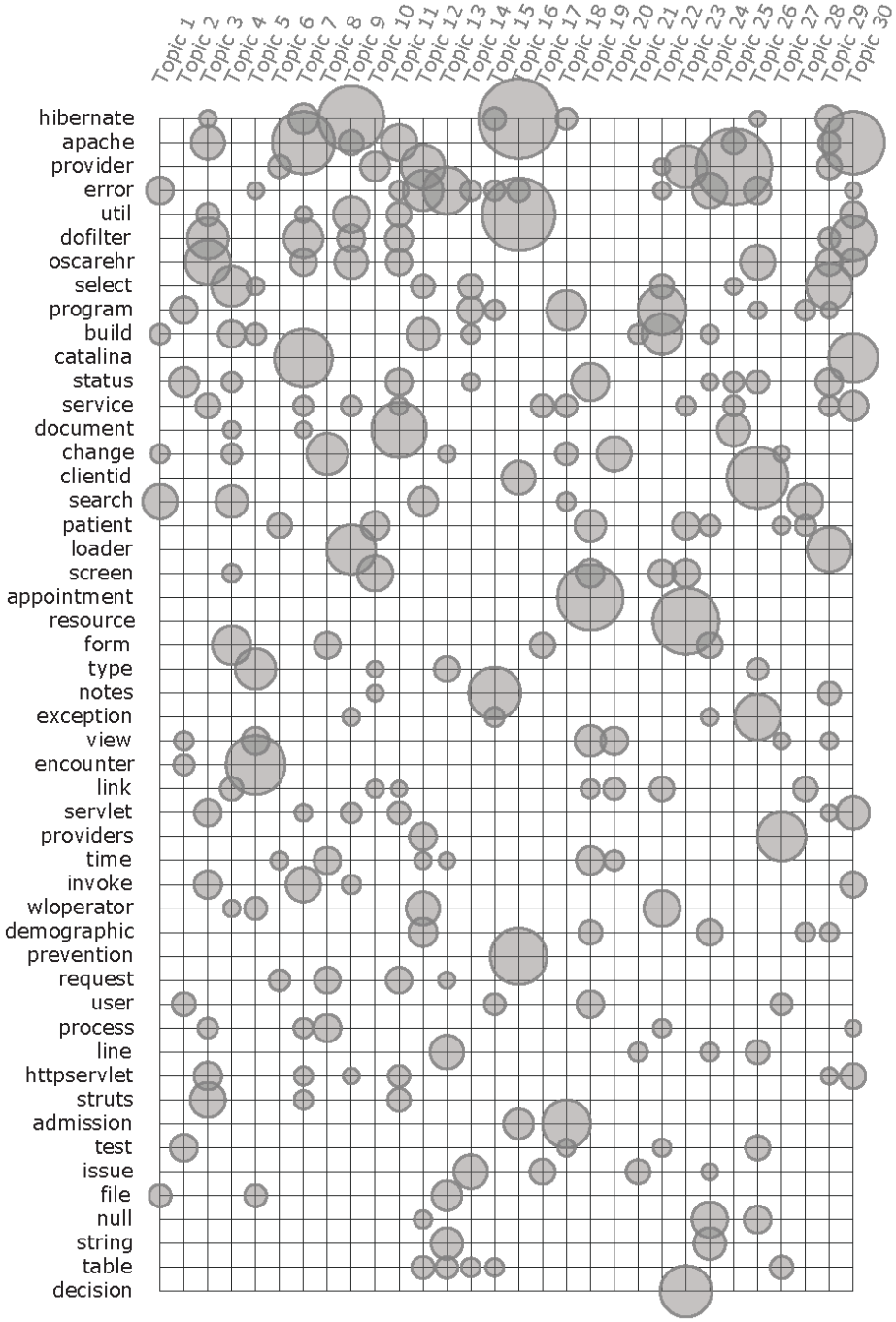

Figure 3: topics extracted from the content of an open-source project's mailing list

ODIN: Interactive Exploration of Concepts and Associations

The first author of this chapter along with collaborators at the University of Georgia was awarded a planning grant from the Institute of Museum and Library Services to develop the Open Data Innovation (ODIN) platform for the goal of making open government data repositories useable by processing and visualizing text data for human sensemaking.[ii] Making complex inter-related information, not just available, but truly accessible, is critical for an informed society. Beyond open government data, ODIN can process any textual data sets and integrate and link various data types which makes it a useful tool for qualitative researchers (Safadi & Boudreau 2017).

Figure 4 shows an interactive session in ODIN in which a researcher explores a large dataset composed of thousands of tweets about data science from 2010 to 2016 (Vaast *et al.* 2015). ODIN offers three integrated views: navigation (left), knowledge (center), and data (left). The navigation view allows researchers to explore data by creating concepts based on search queries. These concepts are then extracted from the text and visualized in the knowledge view as a network in which nodes are entities and linkages correspond to source text entity associations. Interacting with the knowledge view brings the source data into the data view. In this example, the researcher wants to look up tweets related to "big data" and visualize the hashtags and mentions in the tweets. The knowledge view shows a strong association between big data and analytics. The researcher clicks on the link and the tweets are shown in the data view. The researcher can further tune the query, explore the knowledge view by zooming in and out on concepts and relationships, can consult the documents for evidence, annotation, and coding.

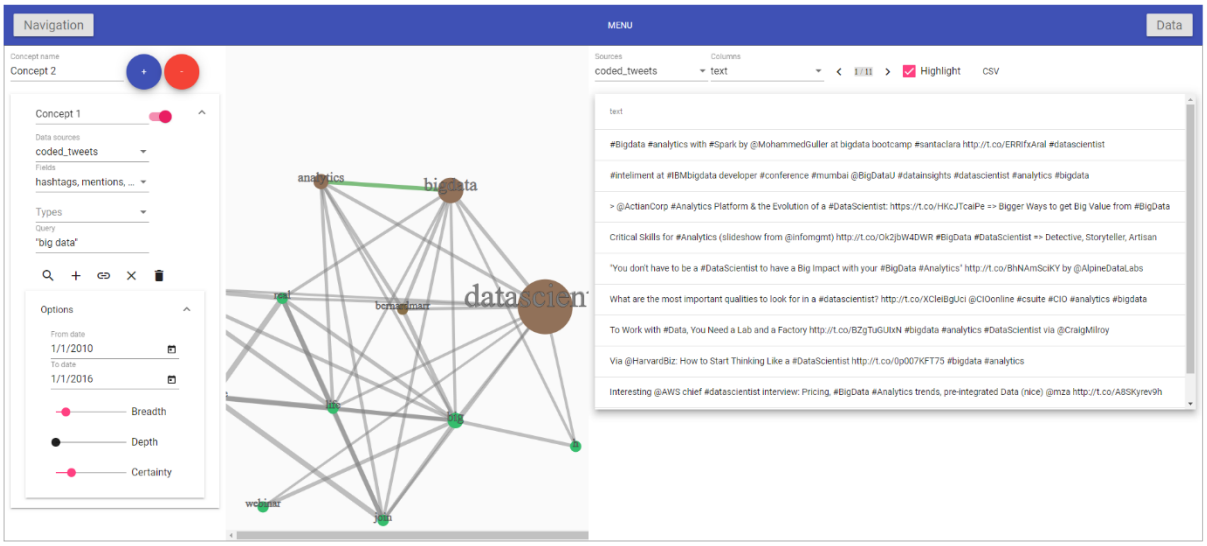

Figure 4: ODIN system in action with a large data set of tweets

The ODIN system allows the researcher to analyze annotated data as well. For example, having coded the data with codes indicating the pragmatics of the tweets and the groups of their tweeters, the researcher can now probe into the association between these two code categories for constant comparison (Figure 5). ODIN reveals commonalities and differences in the pragmatics among groups. For example, students are particularly associated with socializing and discussing skills. Furthermore, ODIN can repeat this analysis overtime to uncover temporal patterns. Finally, researchers can save and share their ODIN sessions with others for collaboration and presentation.

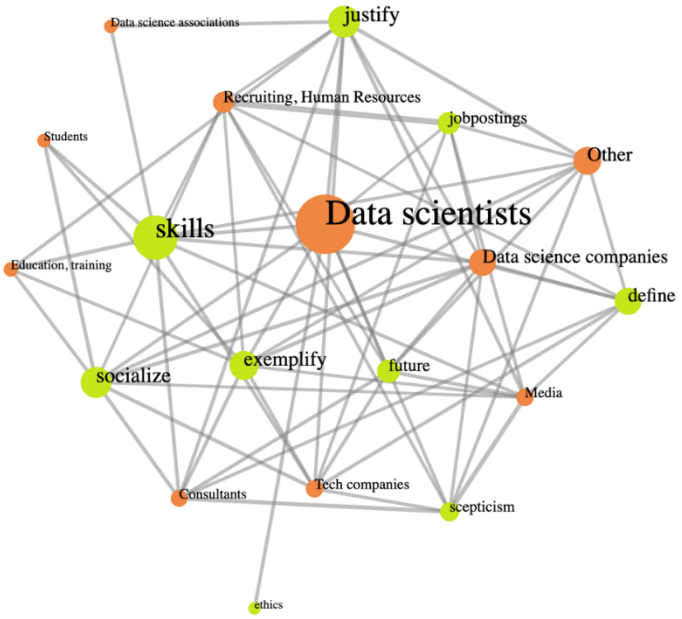

Figure 5: The association among member groups and practices

Under the hood, ODIN uses multiple technologies and algorithms. It relies on open-source libraries for text indexing and processing. It uses information retrieval and named entity recognition for uncovering concepts from the text. It relies on latent semantic analysis and similarity distance to establish associations among these entities. The chosen technologies and algorithms scale for millions of documents and thereby provide efficient and real-time capabilities for analyzing, searching, and querying digital trace data.

Overview: Document Mining and Visualization

Overview is an open-source document mining and visualization system that offers multiple text processing tools and a wide range of visualization and annotation capabilities (Brehmer *et al.* 2014).[iii] Overview is also extensible and enables users to develop their custom visualizations. A unique feature of Overview is processing scanned documents with optical character recognition (OCR) making it a useful tool for researching archived data. Overview supports multiple use cases such as hypothesis generation, hypothesis verification, searching, locating, identifying, and summarization.

Figure 6 shows Overview being used to research 625 White House email messages in scanned PDF formats concerning drilling in the Gulf of Mexico before the 2010 BP oil spill (Brehmer *et al.* 2014). The researcher can probe the documents as well as create multiple visualizations such as the shown word cloud. It is also possible to tag and organize the documents into categories.

CONCLUSION

Social and organizational lives are being digitized, leaving traces that can be used to understand their dynamics and complex nature. Often a roadblock for research, access to data is no longer an issue. On the other hand, researchers have access to multiple new distant contexts in which they analyze more data than was traditionally done before (Abbasi *et al.* 2016; Berente *et al.* 2019; Johnson *et al.* 2019; Levina & Vaast 2015). This new brave digital world creates opportunities and challenges for researchers trained in the qualitative research tradition.

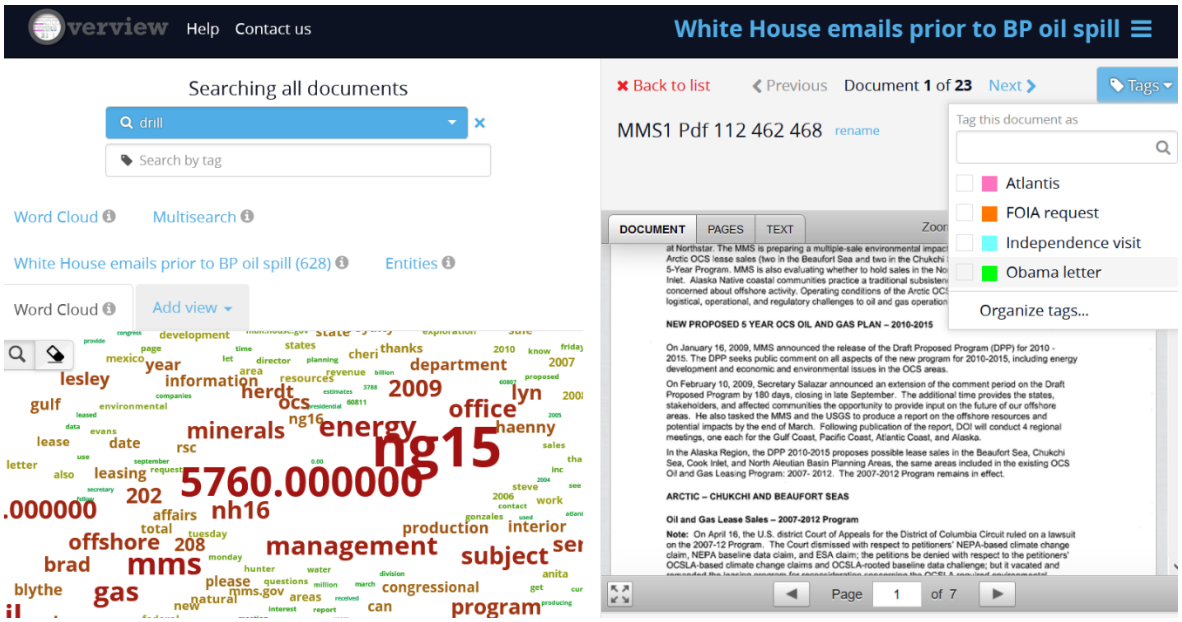

Figure 6: Using Overview to research a collection of scanned documents

In this chapter, we called for expanding the role of visualization to help qualitative researchers in researching digital trace data. Rooted on human information foraging and sensemaking processes, we proposed a paradigm for the continuous interaction between an interactive visualization system capable of extracting and visualizing patterns from textual data, and a researcher whose expertise and interpretive skills are essential for theorizing. We specifically showed how visualization tasks can be used to meet challenges in theoretical sampling, coding, constant comparison, and building a chain of evidence. Finally, we provided a set of criteria to evaluate visualization systems as well as demonstrated three available open-source systems.

Advances in science require advances in methods and tools (Greenwald 2012). Although a readily "off-the-shelf" system is not available yet, we believe that the qualitative research community is not far from achieving this important goal. We hope that this chapter will stimulate further interest within the qualitative research community to come together and invest further effort into creating tools for researching trace data of digitalized phenomena.

Beyond qualitative research methods, our approach builds upon the longstanding discipline of cybernetics (Fararo & McClelland 2006) and underscores the importance of incorporating human deep understanding and sensemaking in data analytics tasks. In the current rush to artificial intelligence, machine learning, cognitive computing, we argue that human judgment is still supreme for ambiguous and challenging tasks (Hindman 2015). Systems designed to leverage large data should provide ample opportunities for human judgment and follow guidelines and design patterns to support this core idea.

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